

MATTA ASHOK VARMA

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SUMMARY

• Android developer with 4+ years building and shipping production **Android** apps using Kotlin, Jetpack Compose, and **Kotlin Multiplatform (KMP) / Compose Multiplatform** — one shared codebase, native UI, end-to-end to Google Play release (latest: [Antar](#) — [Device Info](#)). Proven track record modernizing legacy XML codebases to Compose, eliminating critical OOM crashes, and achieving 99.4% crash-free rates. Expertise in on-device AI (LiteRT, ML Kit, Gemma), hardware integration, Clean Architecture (MVVM/MVI), and AI-assisted development. Delivered 20+ apps across 80+ repositories.

TECHNICAL SKILLS

Programming Languages	: Kotlin, Java, Python, C, SQL, HTML5
Android & Frameworks	: Jetpack Compose, Material 3, KMP, Compose Multiplatform, Koin, Room DB, Coroutines & Flow, ExoPlayer/Media3, Ktor, Coil 3, CameraX, ML Kit, Firebase, WorkManager
Architecture & Patterns	: Clean Architecture, MVVM, MVI, Offline-First, Repository Pattern, Use Cases
AI & Cloud	: Gemma 4 (E2B/E4B), Google LiteRT LM, Gemini API, ComfyUI, Cloudflare R2, Google Maps SDK, HERE Maps SDK
Tools & CI/CD	: Android Studio, Git, GitHub, Gradle, GitHub Actions (CI/CD), Figma, Claude Code, FFmpeg

WORK EXPERIENCE

Android Developer

Feb 2022 – Present

Ramson Softech Pvt Ltd, Hyderabad

- **Architecture & Modernization:** Led UI overhaul of high-traffic Play Store apps, migrating 60% of critical modules from XML to Jetpack Compose. Refactored codebases with Koin DI, reducing boilerplate by 40% and improving maintainability across the product suite.
- **Performance & Stability:** Resolved critical memory leaks and OutOfMemoryError crashes across media services by implementing bitmap recycling, lifecycle-aware coroutines, and safe surface tracking. Achieved **99.4% crash-free rate** and reduced ANR rate by **38%**.
- **On-Device AI in Production:** Shipped real on-device ML features in production apps — ESRGAN super-resolution via **LiteRT** with GPU delegation (incl. TFLite→LiteRT migration for Android 15's 16 KB page-size mandate), **ML Kit** subject segmentation (background removal, portrait blur), and on-device image-labeling + OCR content search. Also built GenAI pipelines (Gemini API + ComfyUI) for server-side processing.
- **Hardware & Data Management:** Designed offline-first Room DB schemas for heterogeneous data types. Engineered hardware-interfacing features for low-latency camera access, flash control, and microphone integration.

Key Production Apps (Live on Google Play Store):

- **GPS Village Maps & Navigation:** Reduced ANR rate by **38%** via GPS callback optimization, cut cold start by **45%** with lazy Koin DI, migrated 60% XML to Compose, integrated Google Maps + HERE Maps dual SDK. DAU increased **2.4x** post-optimization (Firebase Analytics + Play Console).
- **Transparent Live Wallpaper:** Fixed Camera Busy crashes by resolving WallpaperService-Camera2 lifecycle conflicts — **92% crash reduction**. Reduced memory footprint by binding Glide + MediaPlayer to the service lifecycle and improved battery efficiency with frame-skip rendering logic. User retention improved **1.8x** (Firebase Analytics).
- **QR & Barcode Scanner:** Eliminated **100%** of scanner resource leaks via lifecycle-aware coroutines. Scan speed improved **3x** with thumbnail caching. APK size reduced by **22%** removing unused ML model binaries. Crash-free rate improved from 96.1% to **99.4%**.

PROJECTS

Antar — Device Info [\[Play Store\]](#) [\[GitHub\]](#)

- Built and **shipped to Google Play Store, open source on GitHub**: device analytics app with real-time StateFlow streams for 44+ hardware sensors, dual SIM detection, battery health gauges, GPS satellite tracking, and full network deep scan. Custom Compose Canvas charts.
- **Tech Stack:** Kotlin, Jetpack Compose, Custom Canvas, SensorManager, TelephonyManager, Clean Architecture

Chitra: AI Wallpapers [\[Play Store\]](#)

- Built end-to-end from scratch to production and **shipped to Google Play Store**. AI-powered wallpaper app with curated collections, Cloudflare R2 CDN for scalable image delivery, offline-first Room DB caching, and Clean Architecture with MVVM.
- **Tech Stack:** Kotlin, Jetpack Compose, Cloudflare R2, Room DB, Koin, Material 3, MVVM, Clean Architecture

ZeroClawAndroid — AI Agent Daemon [\[GitHub\]](#)

- Engineered an always-on Android foreground service connecting Telegram and WhatsApp (via Twilio) to multiple LLMs with **automatic waterfall failover** and Cloudflare Tunnel for secure remote access.
- **Tech Stack:** Kotlin, Foreground Services, Telegram API, Twilio, Ktor, Cloudflare Tunnel, Multi-LLM Orchestration

MEDHA — On-Device AI Chat Application [\[GitHub\]](#)

- Built a dual-engine AI assistant with **offline mode** (Gemma 4 E2B via Google LiteRT LM — GPU-accelerated on-device text, vision & audio) and **online mode** (Gemini API), enabling private on-device LLM inference with zero data collection. Multi-key API failover, Grand Master AI personas, 22+ quick action templates.
- **Tech Stack:** Kotlin, Jetpack Compose, Google LiteRT LM, Gemma 4, Gemini API, Koin, Room, Material 3, MVVM

Artha — On-Device Multimodal AI (Sense Maker)

- Building a privacy-first app that ingests **audio, video, images, PDFs and docs**, then summarizes, translates and chats with the content **fully on-device** — Gemma LLM engine, ML Kit OCR & translation, TTS/STT pipelines. In active development and daily personal use.
- **Tech Stack:** Kotlin, Jetpack Compose, Gemma (on-device), ML Kit, Koin, Clean Architecture, Material 3

Chitra KMP — Cross-Platform Wallpaper Engine (Android, Desktop, Web) [\[GitHub\]](#)

- Architected a Compose Multiplatform app running on **Android, Desktop (JVM), and Web (Wasm/JS)** from a single Kotlin codebase. Offline-first sync with Cloudflare R2 CDN, 100% shared UI via Voyager navigation, Coil 3 image loading.
- **Tech Stack:** Kotlin, Compose Multiplatform, Ktor, Coil 3, Koin, Voyager, Cloudflare R2, Material 3

ProPlayer — Video Player [\[GitHub\]](#) | WeatherShow KMP — Cross-Platform Weather App [\[GitHub\]](#)

- **ProPlayer:** Modern video player with gesture controls, PiP, subtitles, playlists. 100% Compose + MVVM + ExoPlayer/Media3.
- **WeatherShow KMP:** Cross-platform weather app targeting Android, iOS and Web from one Kotlin Multiplatform codebase with shared business logic. Web deployed to Vercel; iOS build verified on simulator.
- **Tech Stack:** Kotlin, KMP, Compose Multiplatform, ExoPlayer/Media3, Ktor, Koin, Room

EDUCATION

Institute of Aeronautical Engineering

Hyderabad, India

Bachelor of Technology, Electronics & Communication Engineering | 2025